

MITCHELL GRAY

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Education

Georgia Tech, Atlanta, GA

Master's of Science, Computer Science

GPA: 4.0

Artificial Intelligence

Cornell University, Ithaca, NY

Bachelor's of Science, Computer Science

Distributed Systems

Minor, Operations Research Information Engineering

M.Eng Equivalence

Knowledge Domain: Distributed Systems · Systems Programming · AI · Machine Learning · Databases · Cloud

Technical Skills

Languages: C · C++ · Python · Java · Rust · PL/SQL

Software & Tools: Git · Pytorch · Scikit-learn · NumPy · Pandas · Poetry · CMake · CI/CD · Docker · K8s · Azure · AWS

Relevant Work Experience

Oracle

Redwood City, CA

Software Engineer II | C, PL/SQL, Raft, Java, OracleDB, OCI

Sept 2025 – Present

- Working on Raft Replication in the Oracle Globally Distributed Database for Shard Replication · Writing fault-tolerant and scalable Distributed Systems code in **C**, **Java**, and **PL/SQL**.
- Received **Training & Mentorship Excellence Award** · Provided technical mentorship to our newest engineer.
- Designed and implemented a generic shard-operations replication framework that consolidated multiple ad-hoc code paths into a single framework · Improved maintainability, scalability, and ease of creating new operations with fewer defects.
- Improved deployment algorithms when specifying server rack requirements to more evenly spread replicas across shards automatically · Improved CLI user-interface to view user-specified rack information across all shards.
- Implemented comprehensive Pluggable DBs lifecycle management · Enabled seamless Drop, Unplug, and Plug operations of PDBs to streamline maintenance for Raft replicas using Oracle's multi-tenant container architecture.
- Fixed vulnerabilities in slow-replica throttling functionality · Implemented new throttling algorithms to improve fail-over.

Software Engineer I

June 2024 – Sept 2025

- Engineered a fault-tolerant replica replication feature on Oracle Cloud Infrastructure · Enforced placement of Raft replicas on separate server racks to eliminate single-rack replication-unit failures and improve database availability.
- Refactored over 1,200 lines of Sharding serialization code to eliminate a critical data-loss defect · Designed robust de/serialization pipelines and multi-format conversion logic that greatly improved data integrity and system reliability.
- Optimized multiple replica-to-replica data transfer operations · Achieved a **700%** performance improvement without workload and **250%** improvement under active workloads after optimization code changes.
- Resolved critical recoverability issue in replica-to-replica data transfer operations by ensuring complete transfer of essential recovery-related tables and data between replicas · Improved reliability and system integrity during failure.
- Enhanced error-handling mechanisms throughout the codebase · Improved error handling reliability and diagnosability.

Gecko Robotics

Pittsburgh, PA

Software Engineer Intern | C++, Python, C, CMake, GCP, CI/CD

May 2023 – Aug 2023

- Worked on the Robot Controls team · Wrote code for an asynchronous distributed system in **C++**, **Python**, and **C**.
- Revamped Robot & DAQ emulators · Implemented communications protocol and Client/Server TCP networking.
- Replaced VS build-system with **CMake** · Added calibration support to emulators · Integrated **CI/CD** and **Poetry**.

Carnegie Mellon University - Software Engineering Institute

Pittsburgh, PA

DevOps Engineer Intern | Python, Rust, CI/CD, Docker, Kubernetes

May 2022 – May 2023

- Developed **Python**, **Rust**, and **Bash** code on various NDA projects · Updated and created **CI/CD** pipelines · Created GitLab **REST API** live data visualizations for clients · Used ArgoCD and Helm to deploy AWS EKS cluster.
- Built internal analytics tools using **Python**, **Neo4j**, NeoDash, and PageRank-based graph analysis to generate engineering metrics and documentation adopted across multiple teams within the company.

